

Air University



Digital Logic Design (Lab)

Year 2025

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Experiment No. 09

Title: Introduction to Verilog I (Combinational Circuits through Gate-Level Modeling)

Objectives:

- To understand gate-level modeling in Verilog as it relates to combinational circuits.
- To design and implement a half adder circuit using Verilog's basic logic gates (AND and XOR).
- To validate the functionality of the half adder through simulation and truth table comparison.

Equipment and Components:

- Verilog simulation software (e.g., ModelSim, Xilinx Vivado)
- Logic gate primitives: XOR (7486), AND (7408)
- A testbench module for input/output verification

Introduction:

Gate-level modeling in Verilog involves creating digital circuits using essential logic gates such as AND, OR, XOR, and NOT. This low-level representation provides precise control over hardware design, which is key for optimizing performance and resource usage in areas such as arithmetic logic units and data routing. Combinational circuits produce outputs strictly based on current inputs, making them fundamental in digital systems.

In this experiment, we will focus on constructing a half adder, a fundamental combinational circuit that adds two single-bit binary numbers. The half adder yields two outputs:

- **Sum (S):** The result of the binary addition, derived from the XOR of the inputs.
- **Carry (C):** This output indicates if the sum exceeds 1, determined by the AND of the inputs.

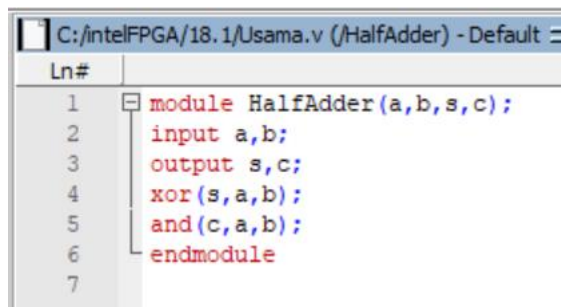
By utilizing Verilog gate primitives, we can effectively model the half adder's behavior and verify its functionality through simulations.

Procedure:

1. Design and Simulation

a. Circuit Design:

- Use an XOR gate to calculate the sum: ($S = A \oplus B$)
- Use an AND gate to determine the carry: ($C = A \cdot B$)

b. Verilog Implementation:

```
C:/intelFPGA/18.1/Usama.v (/HalfAdder) - Default
Ln#
1 module HalfAdder(a,b,s,c);
2   input a,b;
3   output s,c;
4   xor(s,a,b);
5   and(c,a,b);
6   endmodule
7
```

c. Testbench:

- Test all four possible input combinations (00, 01, 10, 11).
- Compare the outputs with the expected truth table using a simulator.

Logic Expressions and Truth Table:**Logic Expressions:**

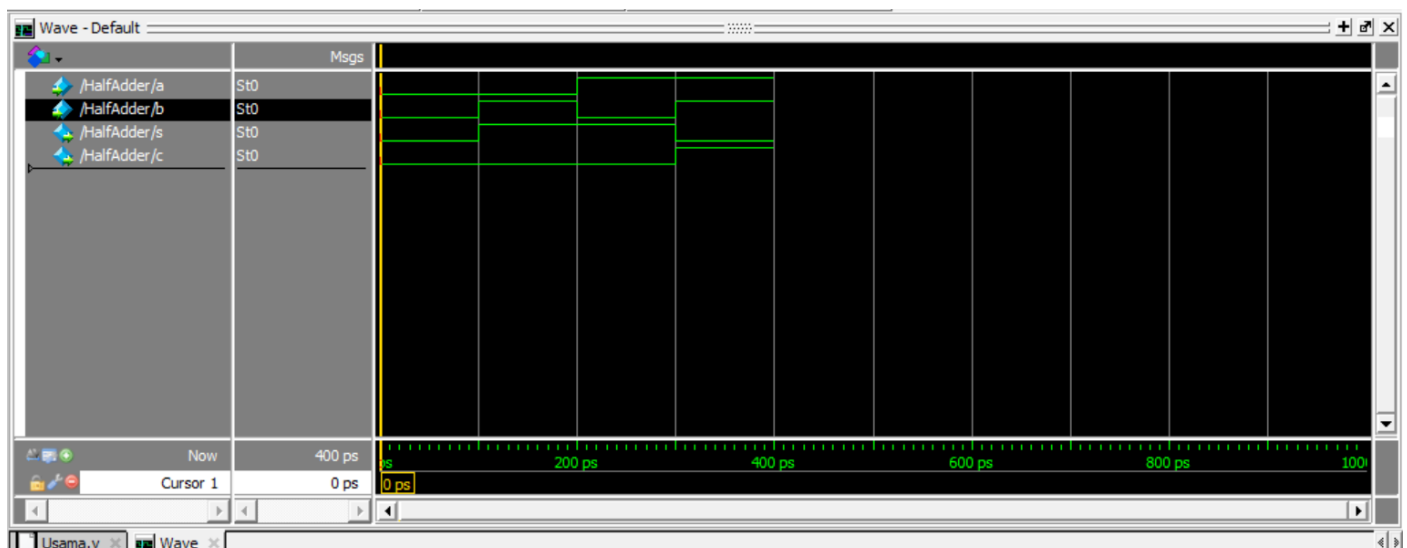
- Sum (S): ($S = A \oplus B$)
- Carry (C): ($C = A \cdot B$)

Truth Table:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Simulated Results:

The simulation results were consistent with the expected truth table, confirming the circuit's accuracy.



Conclusions:

This experiment successfully showcased how to design and simulate a half adder through gate-level modeling in Verilog. The use of XOR and AND gate primitives allowed us to calculate the sum and carry, respectively, with the simulation verifying the circuit's performance. Gate-level modeling provides detailed control over hardware design, which is crucial for optimizing critical paths in digital systems. The half adder serves as a foundational building block for more complex circuits, such as full adders and multi-bit adders.

THE END
